

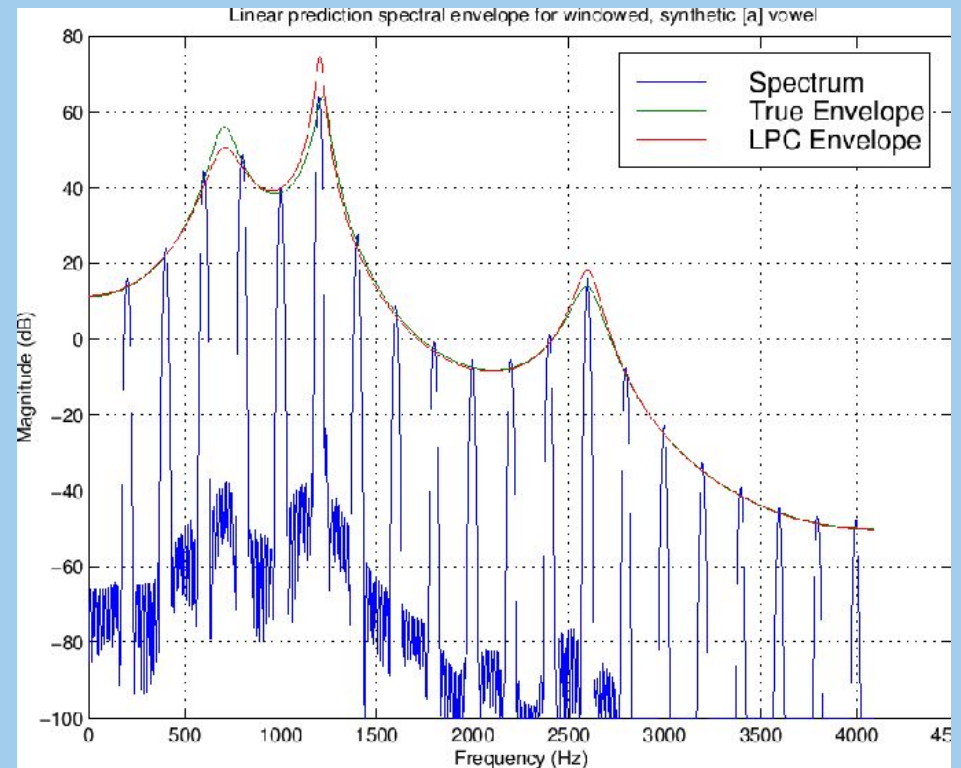
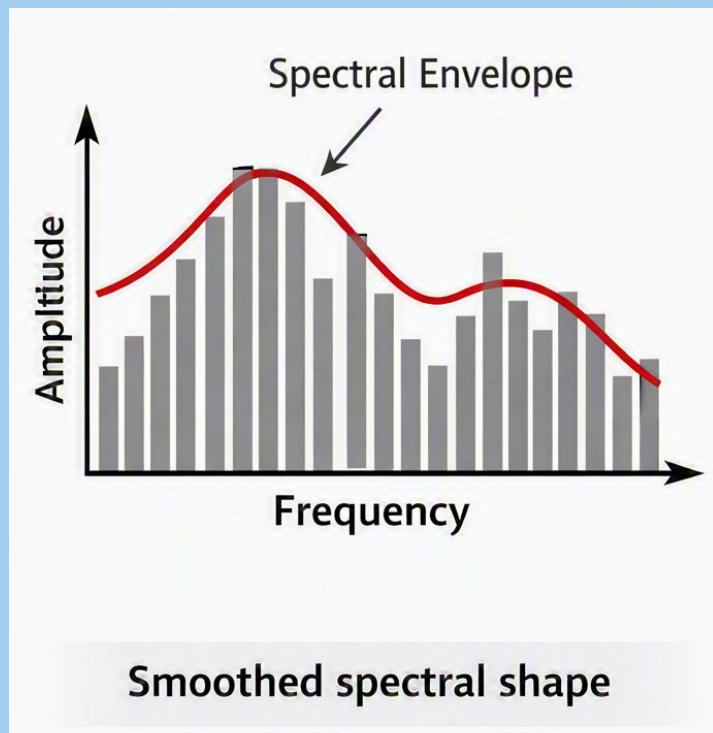
SPECTRUM ANALYSIS MODELS

Reference Text Books:

1. Fundamentals of Speech recognition – L. Rabiner and B. Juang, Prentice Hall signal processing series.
2. Discrete-time speech signal processing: principles and practice, Thomas F. Quatieri, Coth.

SPECTRAL ANALYSIS MODELS

The **spectral envelope** is the smooth curve outlining the peaks and valleys of the speech spectrum, carries crucial information about **phonemes** and **speaker identity**.



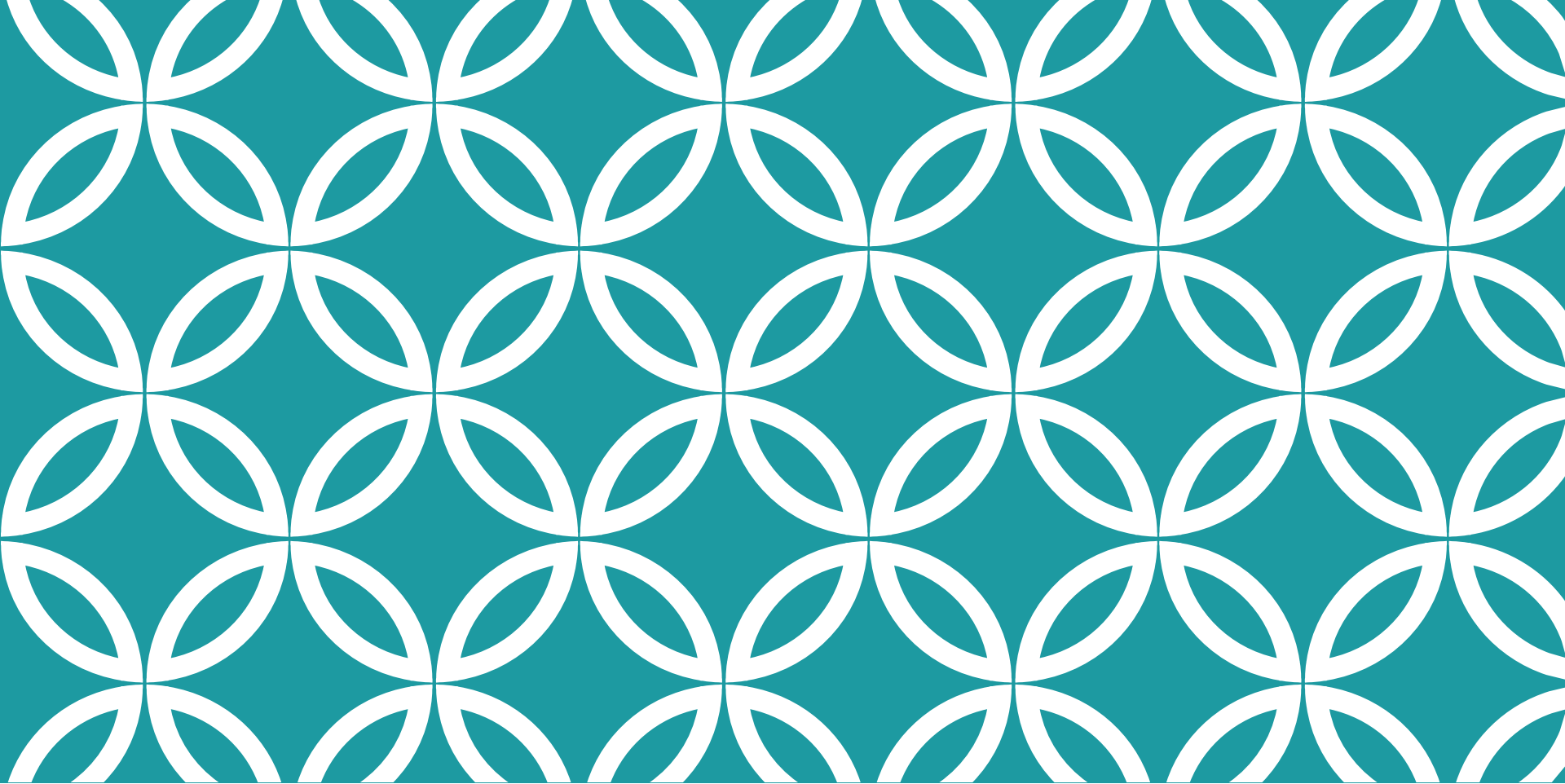
SPECTRAL ANALYSIS MODELS

In speech recognition systems, the signal-processing front end is a crucial component that converts the speech waveform into a parametric representation for further analysis. One of the most important parametric representation is the *short-time spectral envelope*.

Two dominant methods for this are:

- the bank-of-filters model
- the linear predictive coding (LPC) model.

Both methods are used in the initial “parameter measurement” phase of either the pattern recognition or the acoustic-phonetic approach to speech recognition.

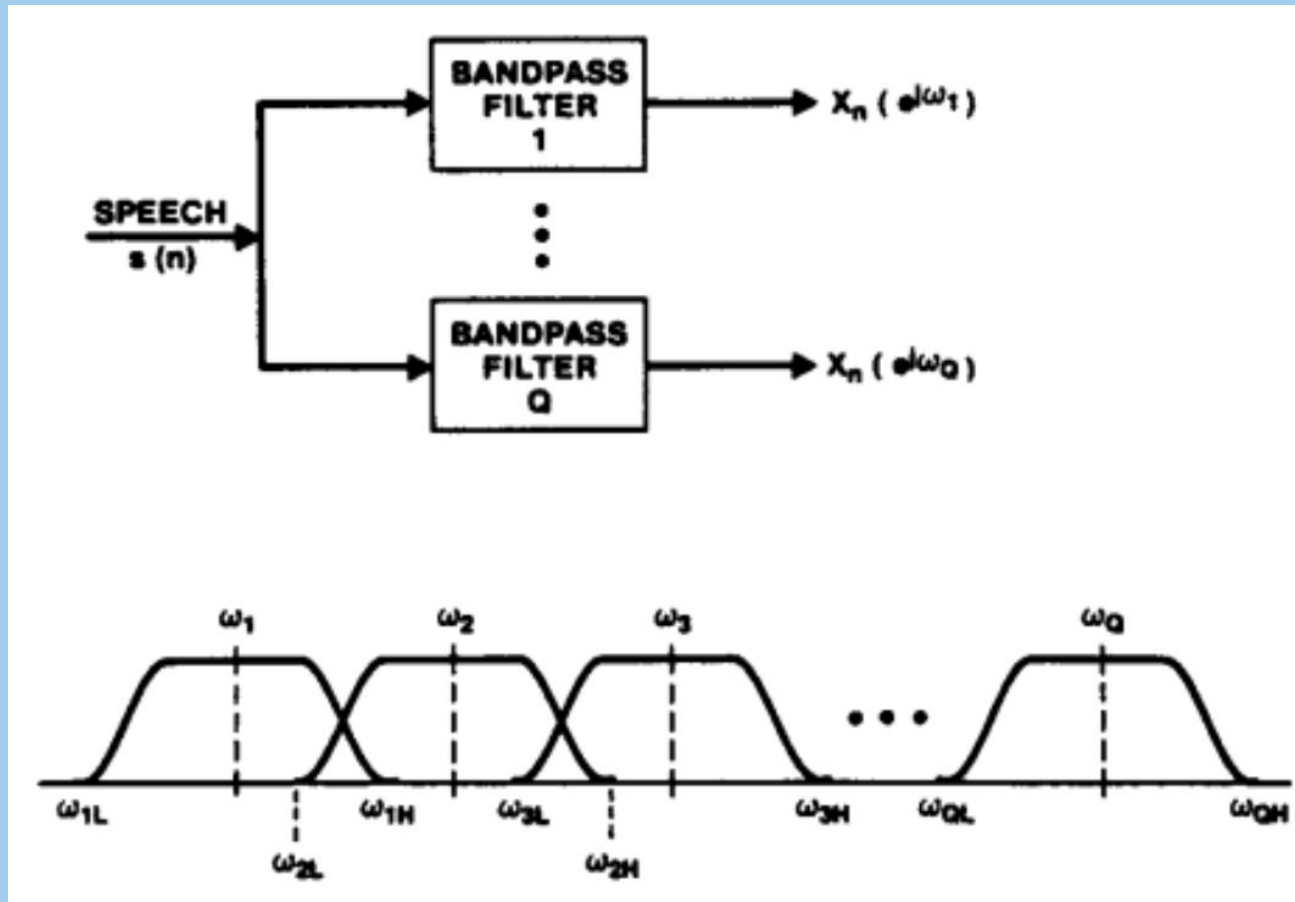


FILTER BANK SPECTRUM ANALYSIS MODEL

Filter Banks
Filter bank Spectrum
Analysis

BANK-OF-FILTERS (BOF) MODEL

The bank-of-filters approach performs spectral analysis by passing the speech signal through a set of bandpass filters that span the frequency range of interest.



TYPES OF DIGITAL FILTERS

□ Uniform Filter Banks:

The filters are spaced uniformly across the frequency range, having equal bandwidth.

The number of filters, Q , should not be too small (less than 8) to resolve the spectrum adequately, nor too large (more than 32) as it might result in some bands having very low energy for certain speakers.

Doesn't match human auditory perception.

□ Non-Uniform Filter Banks:

Filters with varying bandwidths, often narrower at low frequencies and wider at high frequencies

These are designed based on criteria related to human auditory perception.

Complex and computationally heavier than uniform banks.

TYPES OF DIGITAL FILTERS

- ❑ FIR vs. IIR Filters: Filter banks can be implemented using either Infinite Impulse Response (IIR) or Finite Impulse Response (FIR) filters.

FIR filters are

- stable.
- the output is dependent on only current and past inputs,
- can achieve linear phase (waveform shape is preserved), and
- computational cost is significantly reduced using Fast Fourier Transform (FFT) implementations.

A highly efficient way to implement a uniform FIR filter bank is through the STFT. By modulating a fixed lowpass window, the outputs of the filter bank can be computed using an FFT algorithm (polyphase filter-bank structure).

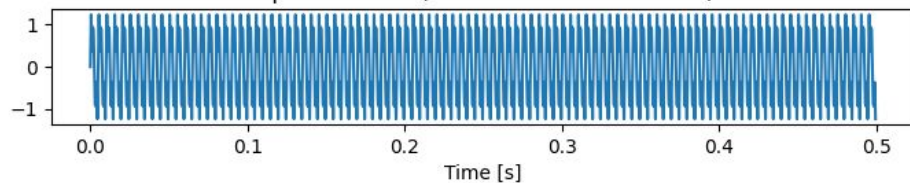
- ❑ Tree Structures:

Non-uniform filter banks, like octave-spaced designs, can be efficiently implemented using a tree structure of Quadrature Mirror Filters (QMFs) with successive stages of filtering and sample rate reduction.

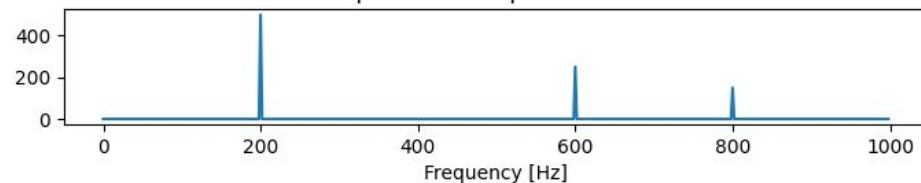
TYPES OF DIGITAL FILTERS

- ❑ Low-pass filter: passes low frequencies, attenuates high.
- ❑ High-pass filter: passes high frequencies, attenuates low.
- ❑ Band-pass filter: passes a specific frequency band.
- ❑ Notch filter: rejects a narrow band, passes others.
- ❑ All-pass filter: passes all frequencies but alters phase.

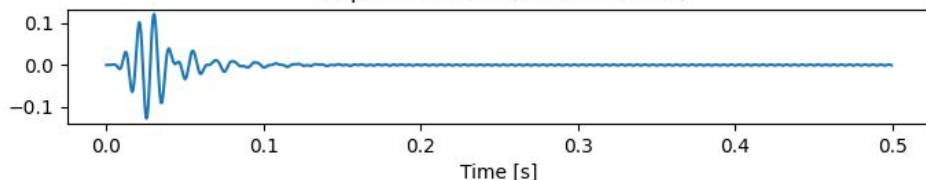
Input Sinusoid (200 Hz+600 Hz+800 Hz)



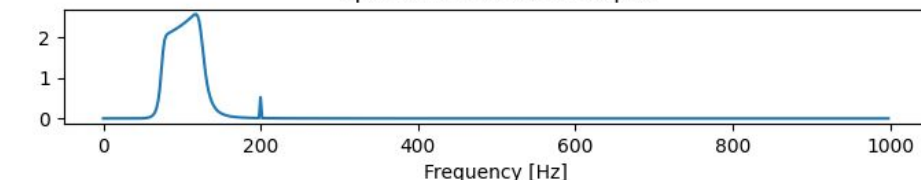
Spectrum of Input Sinusoid



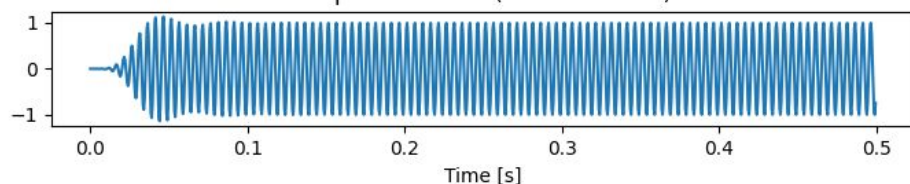
Output of Filter 1 (center 100 Hz)



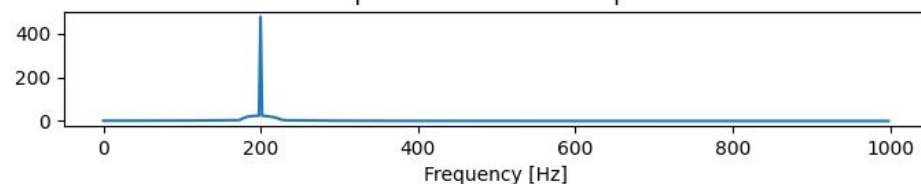
Spectrum of Filter 1 Output



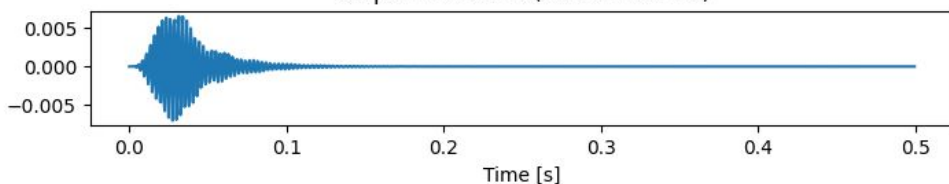
Output of Filter 2 (center 200 Hz)



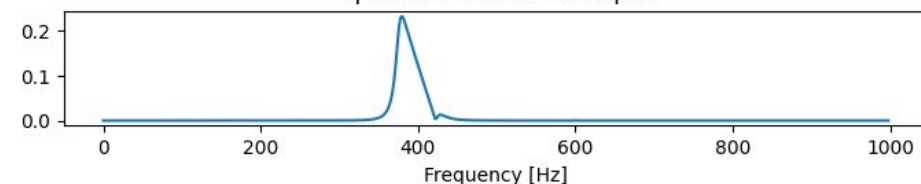
Spectrum of Filter 2 Output



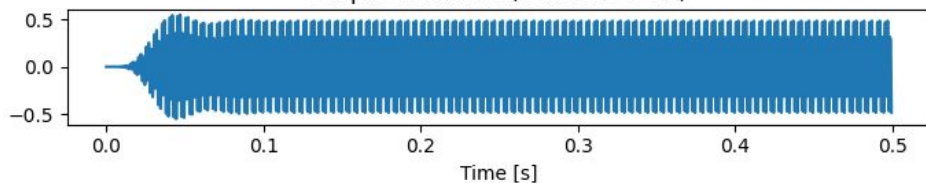
Output of Filter 3 (center 400 Hz)



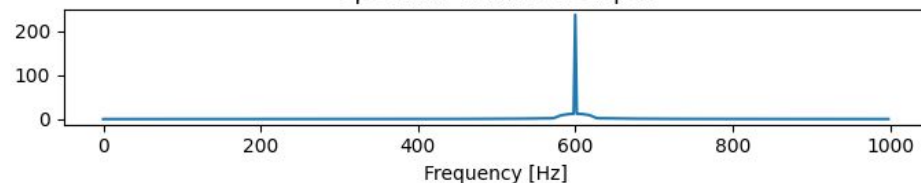
Spectrum of Filter 3 Output



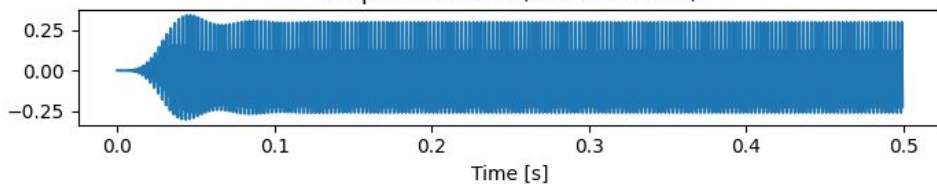
Output of Filter 4 (center 600 Hz)



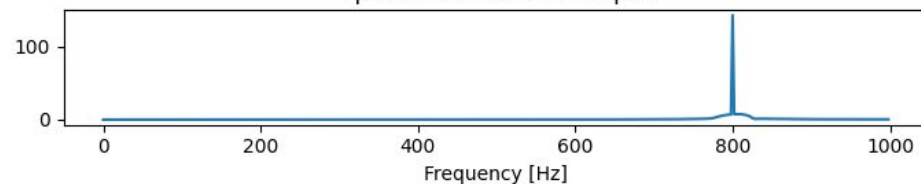
Spectrum of Filter 4 Output



Output of Filter 5 (center 800 Hz)

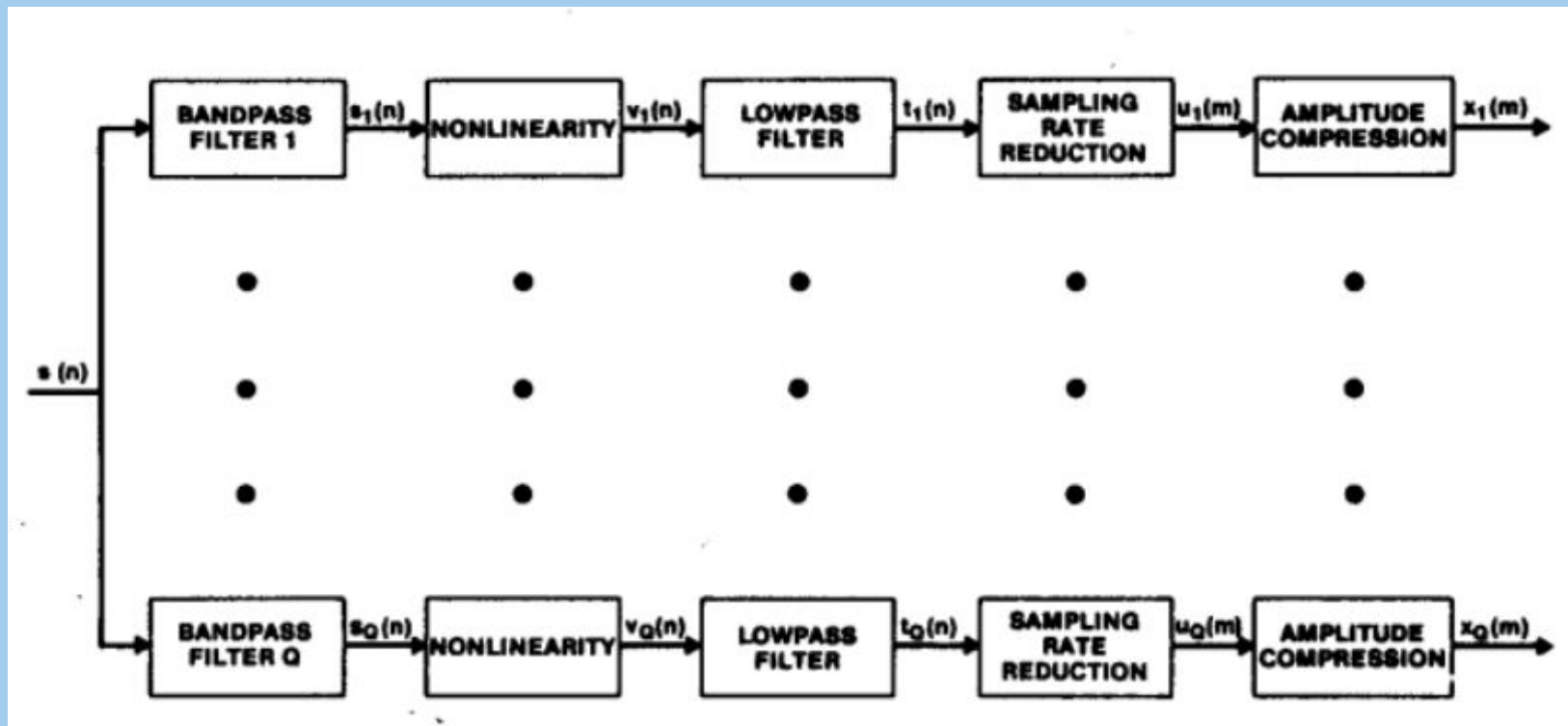


Spectrum of Filter 5 Output



COMPLETE BANK-OF-FILTERS (BOF) MODEL STRUCTURE

A complete bank-of-filters front-end analyzer follows a canonical structure, as shown in Figure.



FILTER BANK SPECTRUM ANALYSIS MODEL

Bandpass Filtering:

The sampled speech signal, $s(n)$, is passed through a bank of Q bandpass filters.

Each filter, $h_i(n)$, has its own impulse response and covers a specific frequency band.

It allows signals within a defined range (between a lower and upper cutoff frequency) to pass.

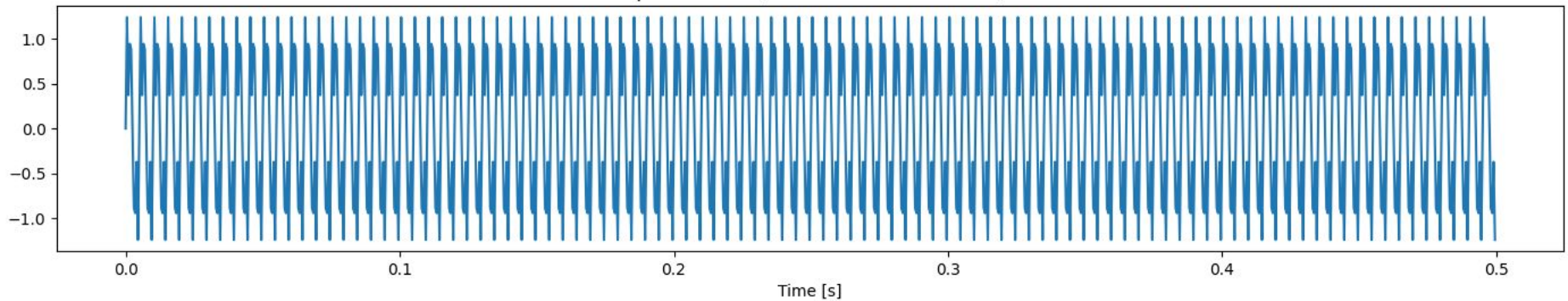
The filters can and generally do overlap in frequency.

The output of the i^{th} filter, a bandpass-filtered signal, is denoted as $s_i(n)$.

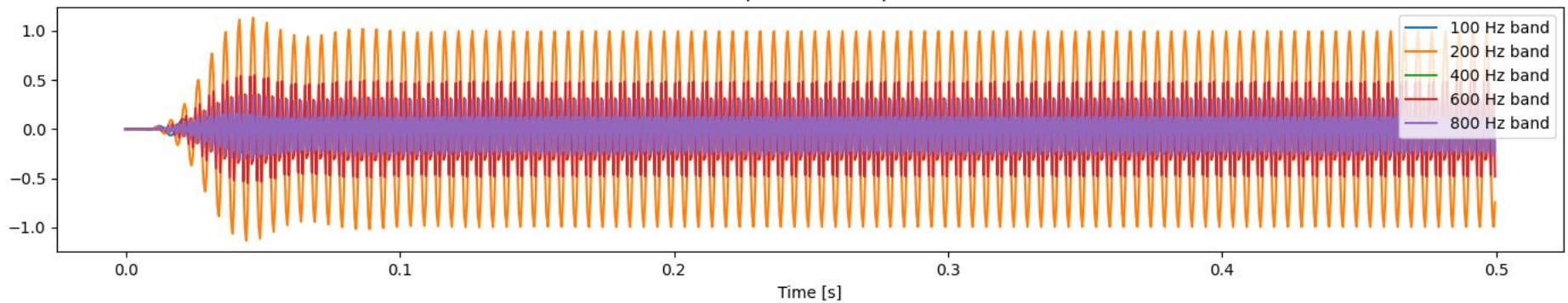
FILTER BANK SPECTRUM ANALYSIS MODEL

Bandpass Filtering:

Input Sinusoid (200 Hz+600 Hz+800 Hz)



Bandpass Filter Outputs



FILTER BANK SPECTRUM ANALYSIS MODEL

Nonlinearity (Rectification):

To measure the energy in each band, the bandpass signals $s_i(n)$ are next passed through a nonlinearity, typically a full-wave or half-wave rectifier.

This process shifts the signal's spectrum to the low-frequency band while creating high-frequency images.

Full-wave rectifier takes the absolute value of the signal, flipping negative parts positive.

Half-wave rectifier sets negative parts to zero, keeping only the positive half.

FILTER BANK SPECTRUM ANALYSIS MODEL

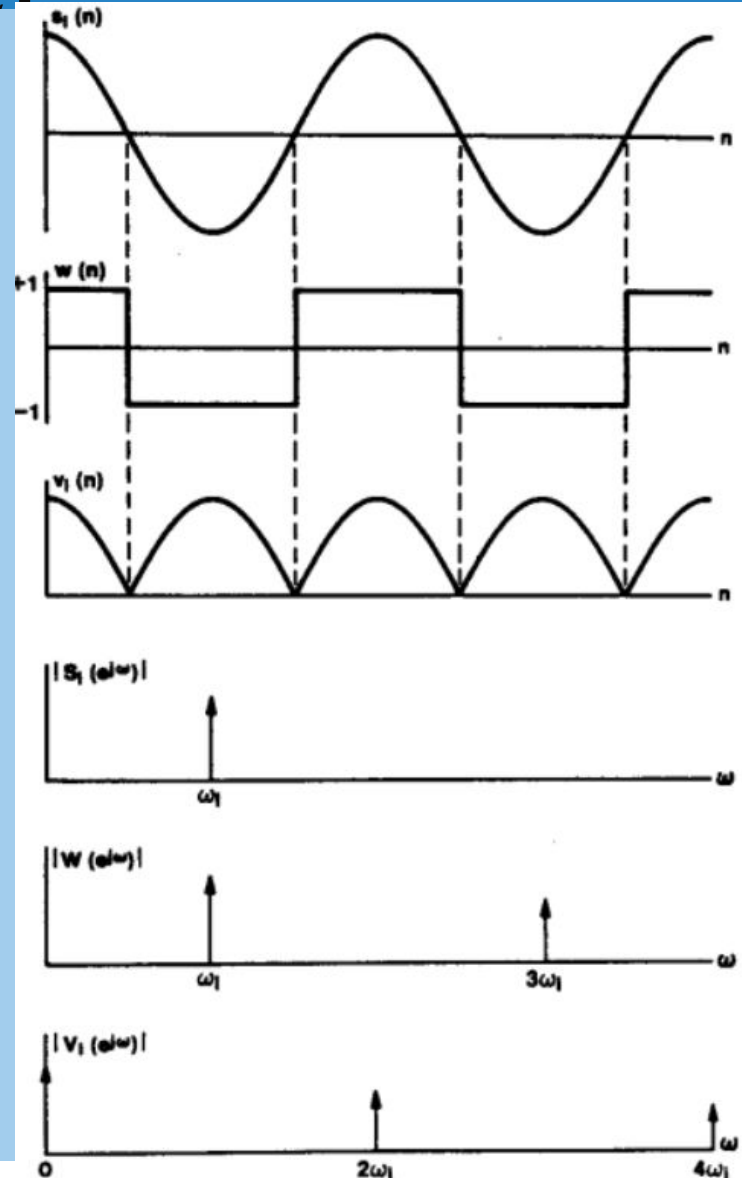
Nonlinearity (Rectification)

Rectification produces:

High-frequency harmonics/images ($2f$, $4f$, $6f$...) that are artifacts.

A DC component (constant offset at 0 Hz).

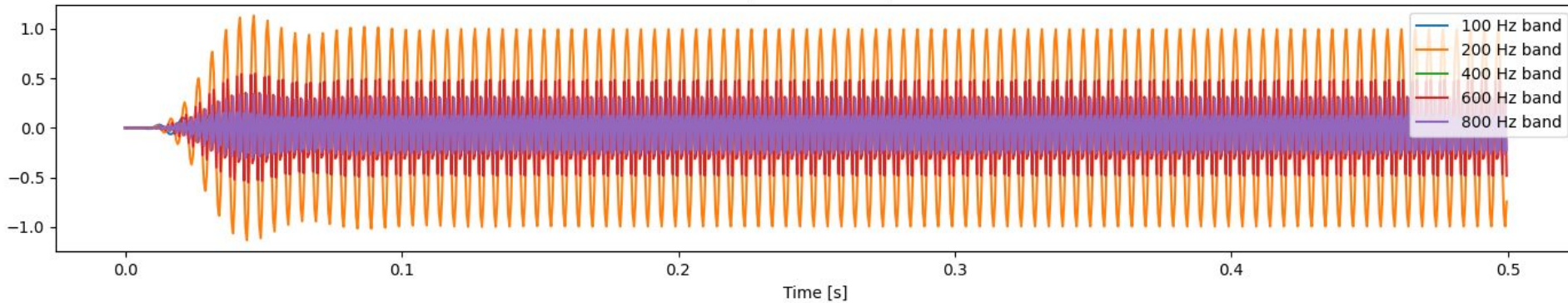
Low-frequency envelope terms (slow variations that track energy).



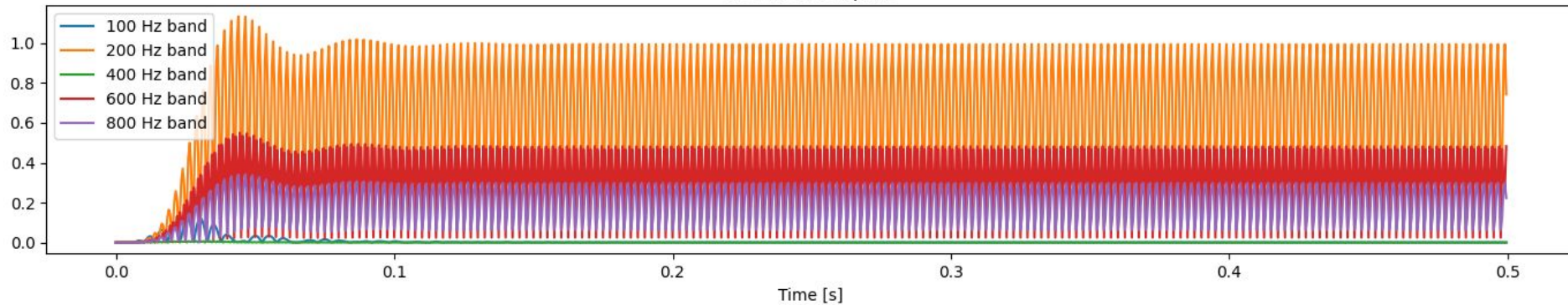
FILTER BANK SPECTRUM ANALYSIS MODEL

Nonlinearity (Rectification):

Bandpass Filter Outputs



Rectified Outputs

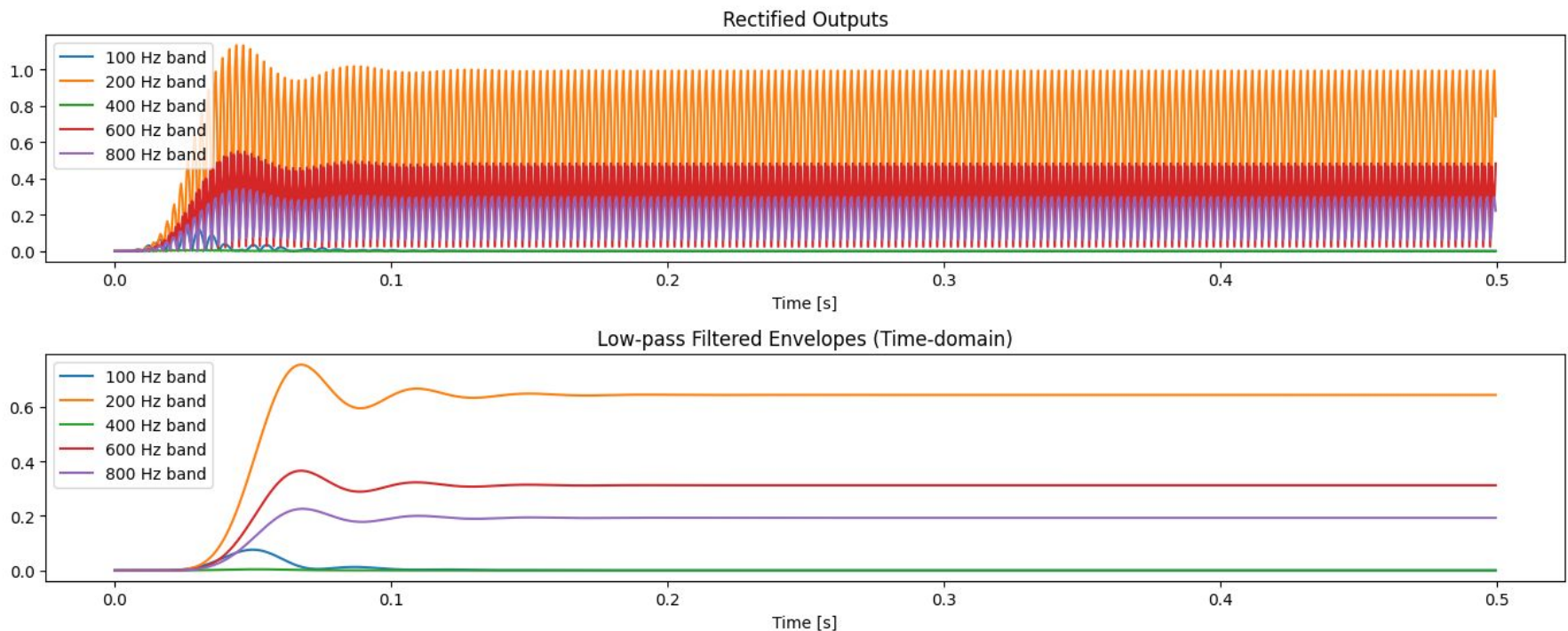


FILTER BANK SPECTRUM ANALYSIS MODEL

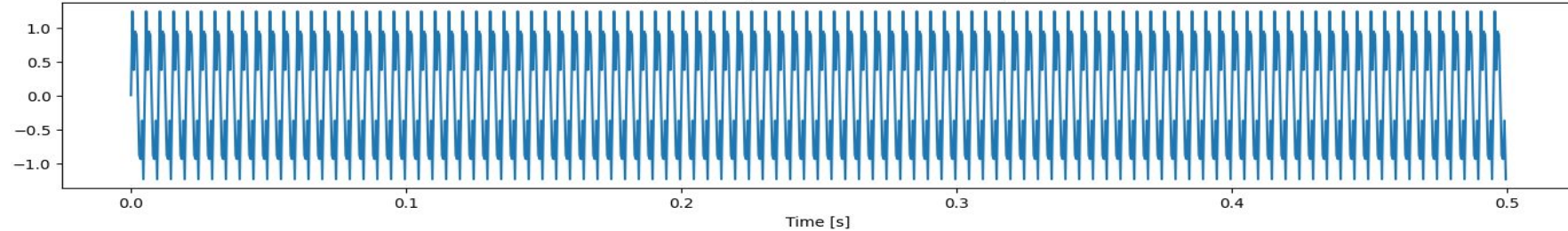
Lowpass Filtering:

A lowpass filter is applied to the output of the nonlinearity to smooth the signal and eliminate the newly created high-frequency images, retaining the DC component that represents an estimate of the speech signal energy in that frequency band.

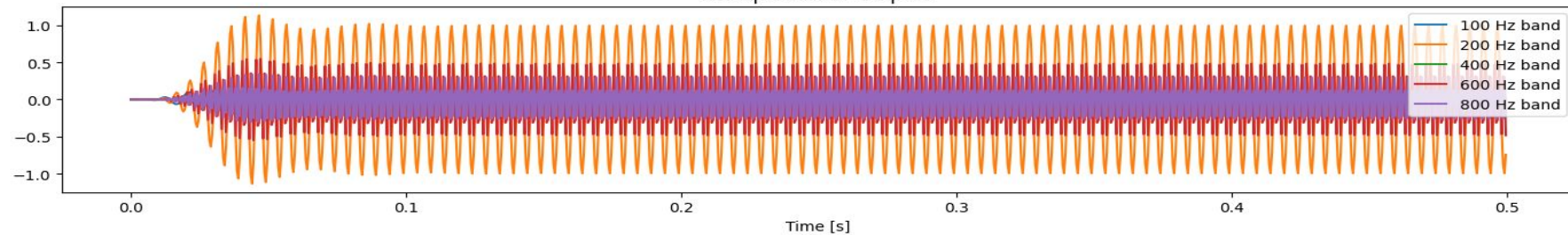
For quasi-periodic speech, this results in energy concentrated in a low-frequency band around DC.



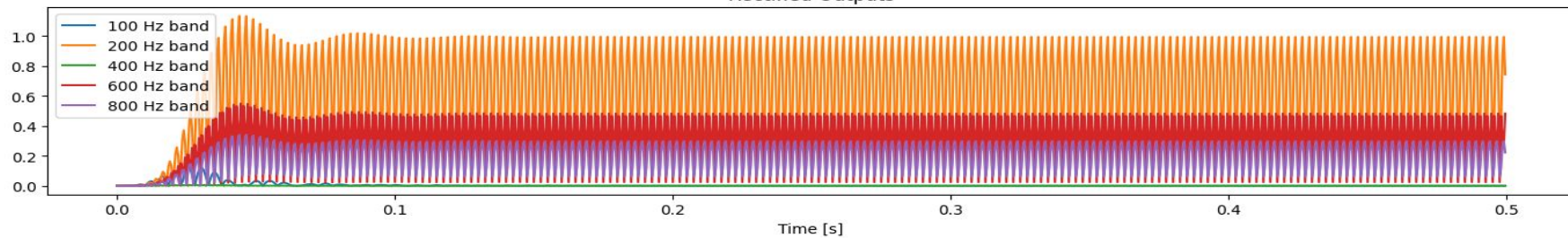
Input Signal (200 Hz + 600 Hz + 800 Hz)



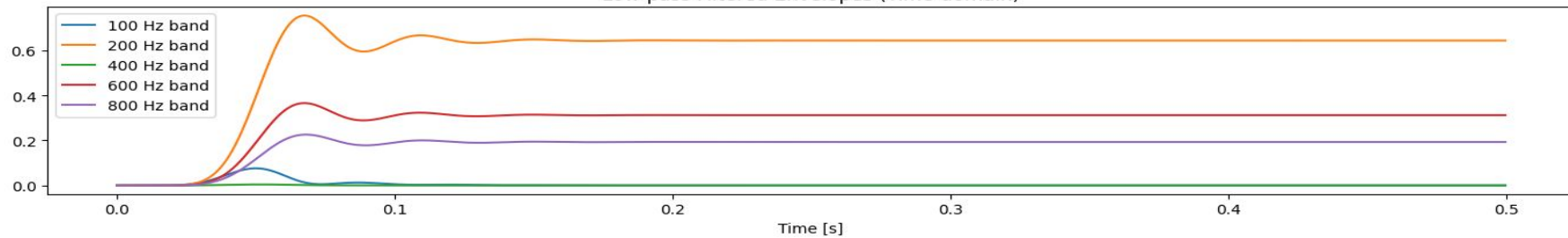
Bandpass Filter Outputs



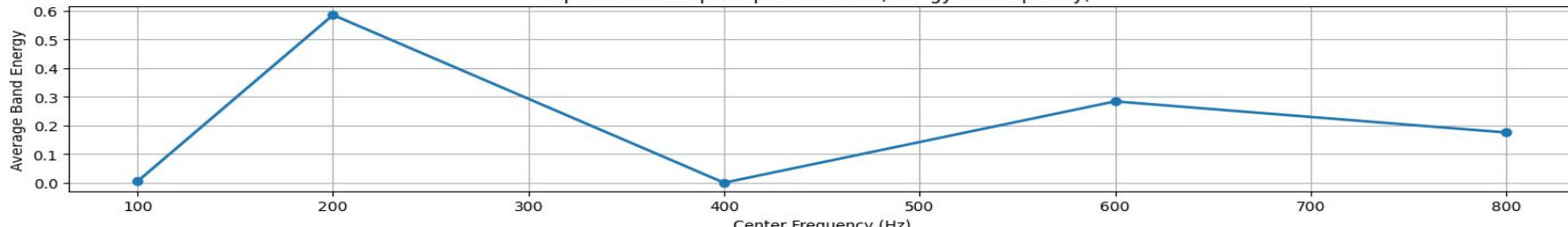
Rectified Outputs



Low-pass Filtered Envelopes (Time-domain)



Spectral Envelope Representation (Energy vs Frequency)



FILTER BANK SPECTRUM ANALYSIS MODEL

Sampling Rate Reduction (Decimation): The bandwidth of the resulting lowpass signal $u_i(n)$ is related to the rate of change of speech harmonics and is typically around 20–30 Hz. Therefore, for efficiency, the signal is resampled at a much lower rate, on the order of 40–60 Hz.

Amplitude Compression: The dynamic range of the signal is compressed using a logarithmic or μ -law encoding scheme.

This complete process significantly reduces the information rate of the speech signal.

For example, a wideband speech signal sampled at 20 kHz with 12-bit precision has a raw bit rate of 240 kbits/sec. A 16-channel filter bank output, sampled at 50 Hz with 7-bit compression, reduces this to 5600 bits/sec, achieving roughly a 40-to-1 data reduction.

FILTER BANK SPECTRUM ANALYSIS MODEL

Exercise Question:

A narrowband speech signal is sampled at 8 kHz with 8-bit precision, giving a raw bit rate of 64 kbits/sec. Suppose this signal is processed through a 10-channel filter bank, where each channel output is sampled at 100 Hz and quantized with 6-bit precision.

Compute the raw bit rate of the original signal.

Compute the compressed bit rate after filter bank processing.

Determine the compression ratio achieved.

NON-UNIFORM FILTERS - MEL FILTER

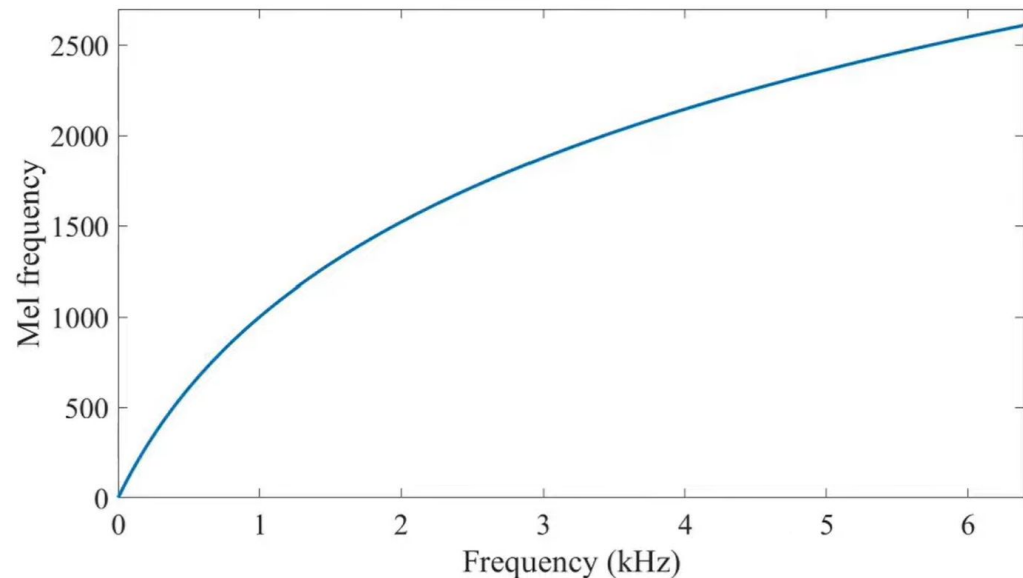
"Humans perceive frequency logarithmically"

Filters can be spaced according to the **critical band scale** (or related scales like the **mel scale** or **bark scale**), based on perceptual studies of how different frequency bands contribute to speech articulation.

Mel-scale:

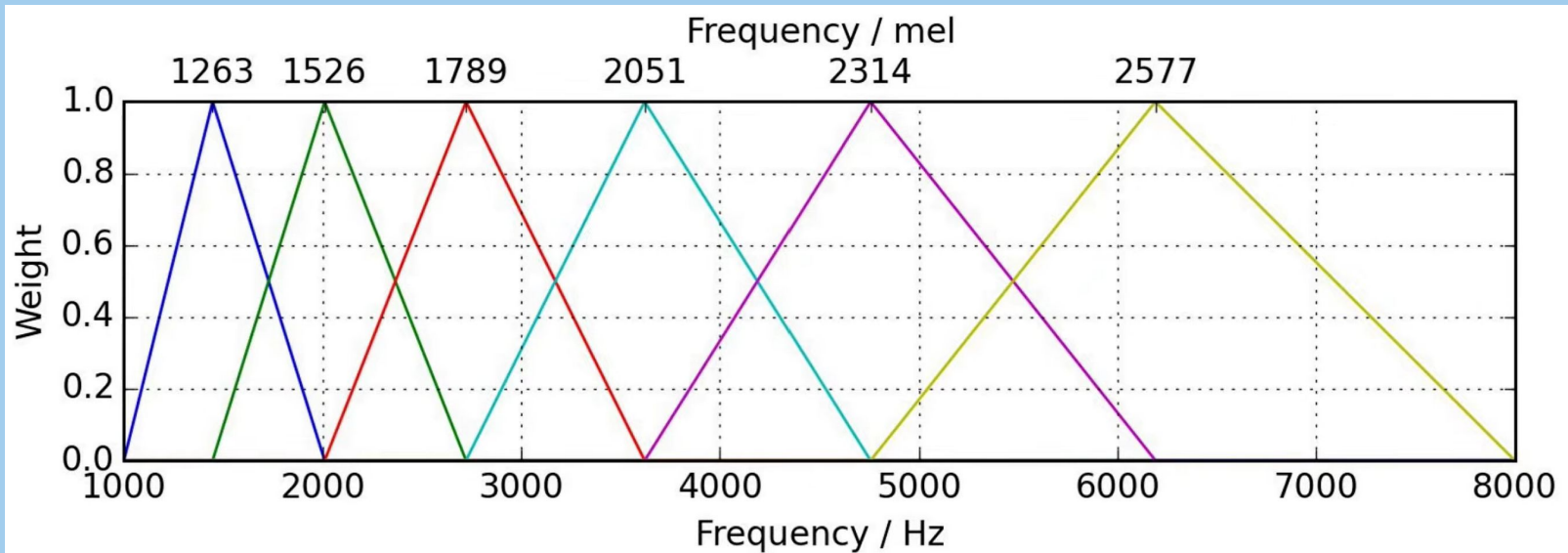
$$m = M(f) = 2595 \cdot \log_{10}\left(1 + \frac{f}{700}\right), \quad f = 700\left(10^{\frac{m}{2595}} - 1\right)$$

- Nearly linear spacing for low frequencies
- Logarithmic spacing for high frequencies.



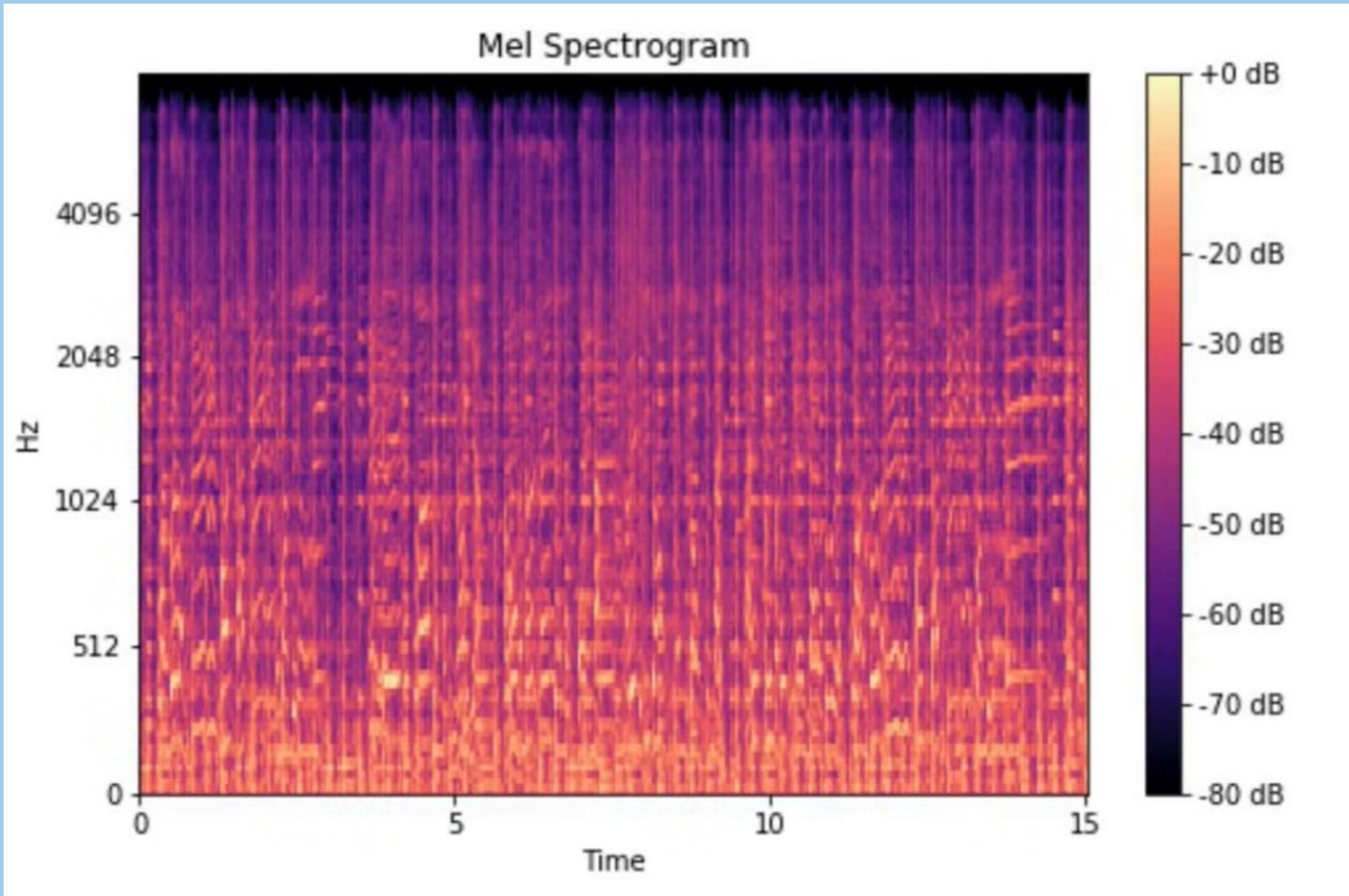
MEL FILTER BANK

1. Convert the lowest and the highest frequency to Mel
2. Create the bands with equally spaced points
3. Convert points back to Herz
4. Approximate the frequency bins and create triangular filters



MEL SPECTROGRAM

After application of Mel Filter Bank, Mel Spectrogram is generated.



APPLICATIONS OF FILTER BANK MODEL

Speech Recognition

- Mel filters are used to extract features like MFCC for automatic speech recognition.

Speaker Identification

- Filter banks preserve spectral envelope details that distinguish individual voices.

Source Separation

- Cocktail Problem: Applied in separating overlapping voices or instruments in audio recordings.

Speech Enhancement

- Spatial filter banks help in multichannel speech enhancement, making speech clearer in noisy environments.

Speech Synthesis

- Filter banks can approximate the resonances of the vocal tract (formants).
- Modern systems (e.g., Tacotron, WaveNet) often use Mel-spectrograms derived from filter banks as input representations.

Musical Analysis

- Filter banks extract timbre and harmonic features for tasks like instrument recognition, genre classification, and music transcription.